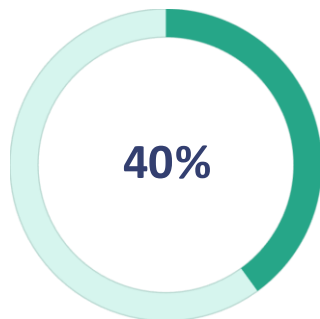


AI in Education

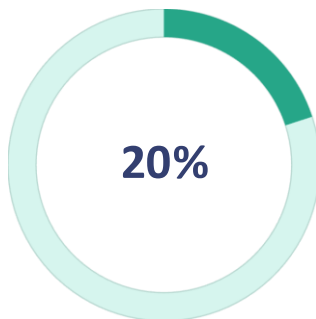
A 20-Hour Interactive Course

A comprehensive professional development program divided into three modules, designed to equip educators with practical AI skills for teaching, research, and innovative pedagogy.



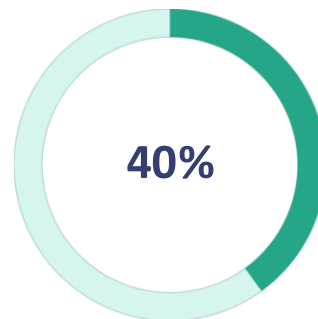
AI in Teaching

8 hours



Ludopedagogy

4 hours



AI in Research

8 hours

3 Modules

20 Hours Total

Interactive Learning

Module 1

AI in Teaching

1

Session 1

Introduction to AI & Content Generation.

2

Session 2

Prompt Engineering — writing prompts that deliver results.

3

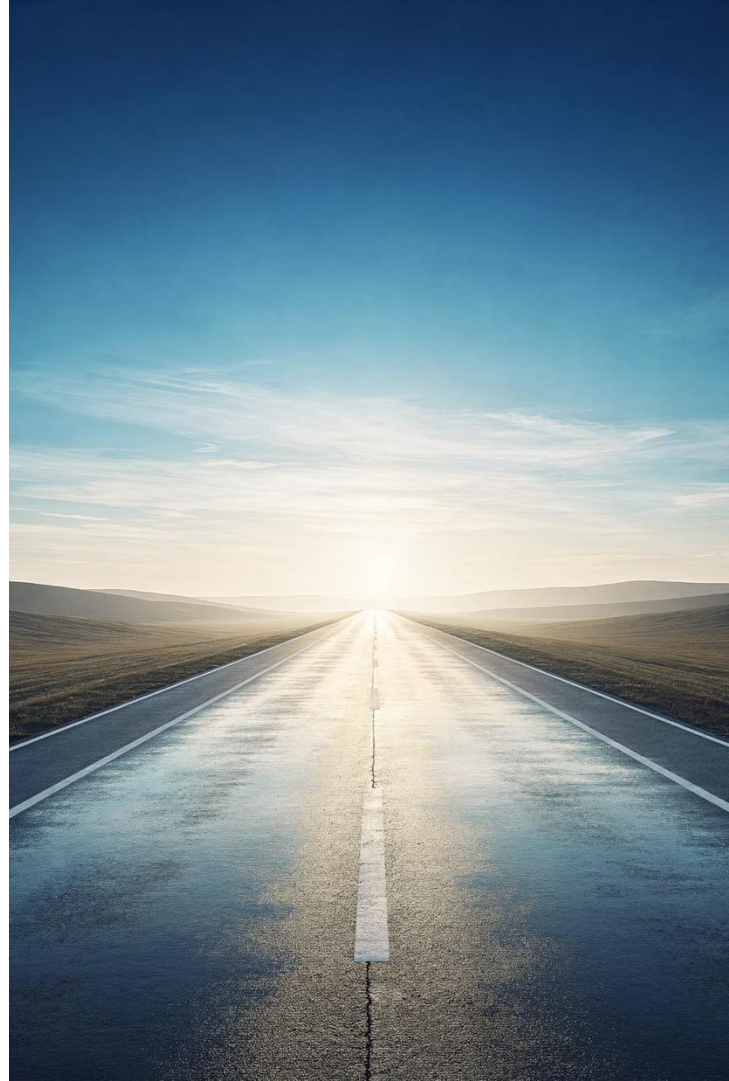
Session 3

AI as a Teaching and Learning Assistant

4

Session 4

Ethical Issues, Responsibilities, and Learning Designer



Module 1

AI in Teaching

1

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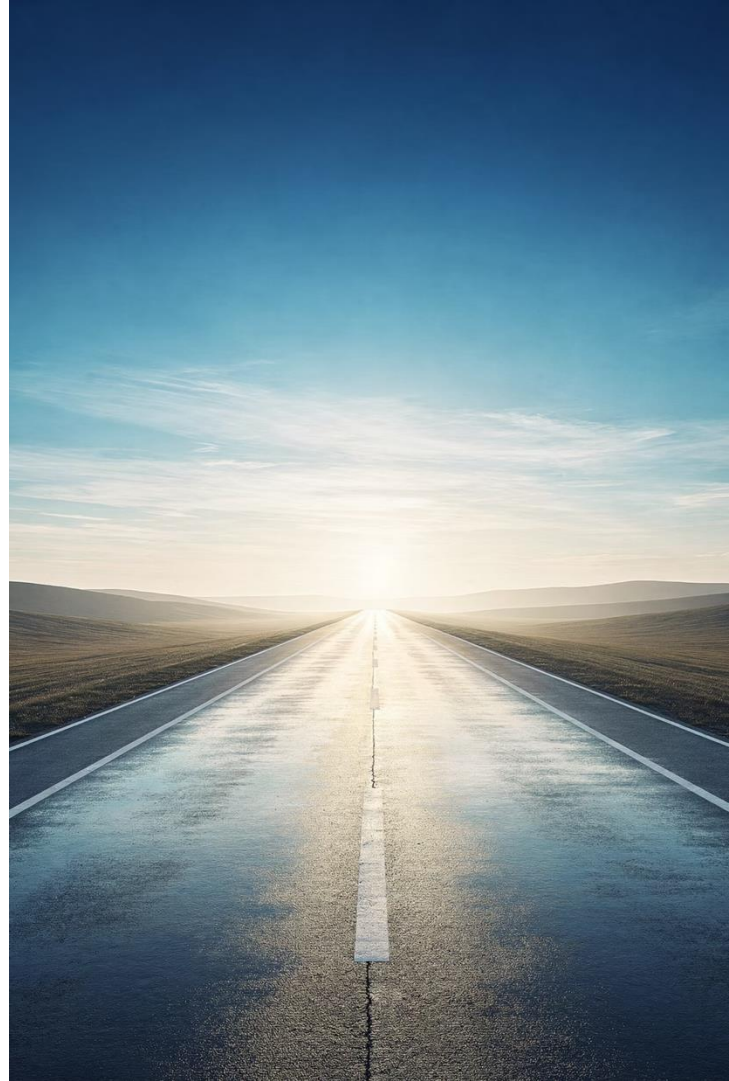
Session 3

AI as a Teaching and Learning Assistant

4

Session 4

Ethical Issues, Responsibilities, and Learning Designer





Session 1

Introduction to AI & Content Generation

Embark on our first session to explore the foundational concepts of Artificial Intelligence and its transformative role in creating diverse educational content.

01

Unpacking AI Fundamentals

Grasp the core definitions and differentiators of AI, Machine Learning, Deep Learning, and Natural Language Processing.

02

Exploring Content Generation

Discover how AI algorithms produce text, images, and other multimedia for teaching and learning.

03

Identifying Key Limitations

Understand the current boundaries and ethical considerations of AI tools in educational contexts.

Session Objectives

By the end of this session, you will have a clear foundation to start integrating AI into your teaching practice. This is about **understanding and discovering** — not mastering every tool overnight.



Understand AI

Build a practical, jargon-free understanding of what artificial intelligence is and how it works.



Discover Key Tools

Explore the leading AI tools educators are using today, from text and slides generation to images and video creation.



See AI in Teaching

Understand the concrete ways AI can support lesson planning, content creation, and student engagement.



Create Your First Content

Write your first AI prompt and generate a real piece of teaching material before the session ends.



Let's Start With You

How are you currently using AI?



Not at All

Haven't tried any AI tools yet — this session is your starting point.



Occasionally

Tried a tool or two, but haven't made it part of your regular practice.



Regularly

Already using AI tools in your workflow — ready to go deeper.

📖 KNOWLEDGE 🗣️ TEACHING USE 📊 EVALUATION ⚡ AI TOOLS

Your Experience with AI

Let's assess your experience

Connect to Mentimeter and answer a few quick questions about your current interaction with AI.

- No right or wrong answers – this is a learning journey.
- Your responses will help us adapt the session to better suit everyone's needs.

Visit [menti.com](https://www.menti.com)

Enter the code 7246 7377



menti.com

7246 7377

Why AI Matters in Education

AI is already reshaping education — from how we teach to how students learn.



How We Teach

AI assists with lesson planning, differentiation, and personalised feedback at scale.



How Students Learn

Instant explanations, interactive tutoring, and self-paced content are now possible.



How Content Is Created

Slides, quizzes, handouts, and videos can be drafted in minutes, not hours.

The question is not whether to engage with AI — it is how to engage with it wisely.

What Is Artificial Intelligence?

Artificial Intelligence refers to computer systems that can perform tasks which normally require human-like thinking. You do not need a technical background to use these tools effectively — but a basic understanding helps you use them better.



Understanding Language

AI can read, interpret, and respond to text and speech, making conversation-based tools possible.



Solving Problems

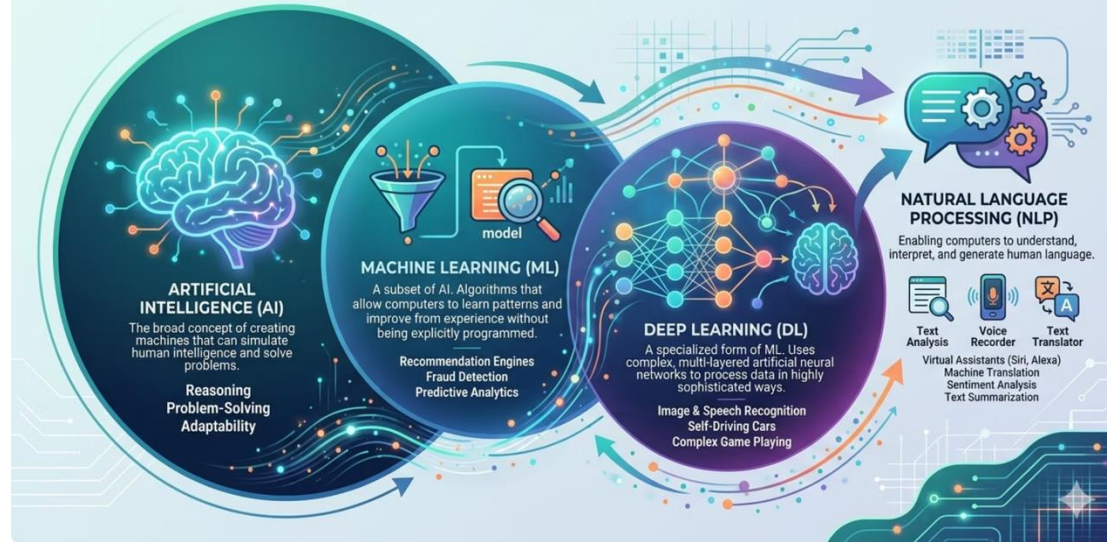
AI can analyse data, recognise patterns, and suggest solutions — from scheduling to diagnosing errors.



Generating Content

AI can produce original text, images, code, and audio based on a simple instruction from you.

INTRODUCTION TO ARTIFICIAL INTELLIGENCE



Traditional Software vs. AI

Understanding this distinction helps explain why AI behaves differently from the apps you already use — and why that unpredictability is both a strength and a challenge.

Traditional Software

- **Fixed rules:** Follows explicit instructions written by a programmer.
- **Deterministic:** Same input always produces the same output.
- **Predictable:** Reliable, but limited to what it was programmed to do.



Artificial Intelligence

- **Learns patterns:** Trained on vast amounts of data, not hand-coded rules.
- **Probabilistic:** Generates the most likely response based on context.
- **Variable output:** May respond differently to the same question each time.

- 📌 Key insight: AI does not "follow rules" — it **predicts** the most probable, helpful response. This is why phrasing your prompts well makes such a big difference.

Analytical AI vs. Generative AI

There are two broad families of AI tools. In this session — and throughout this series — we focus on **Generative AI**, which is the type that creates new content from your instructions.



Analytical AI

What it does: Analyses existing data to find patterns, make predictions, or flag anomalies.

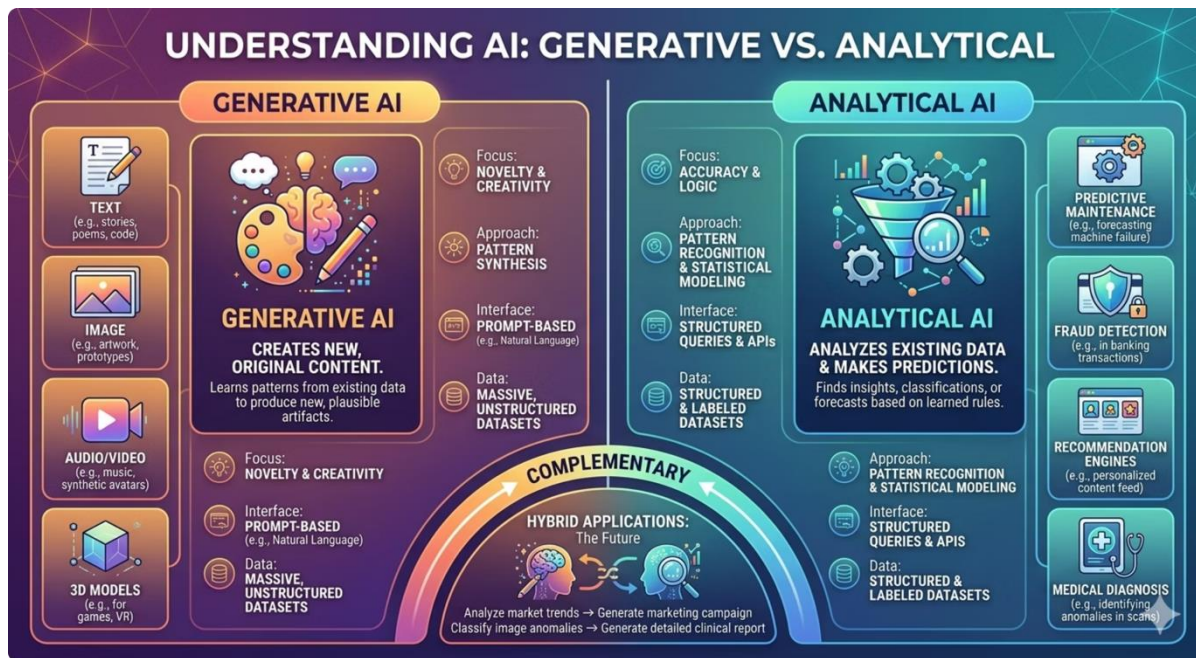
Examples: Spam filters, recommendation engines, plagiarism detectors, student performance dashboards.



Generative AI

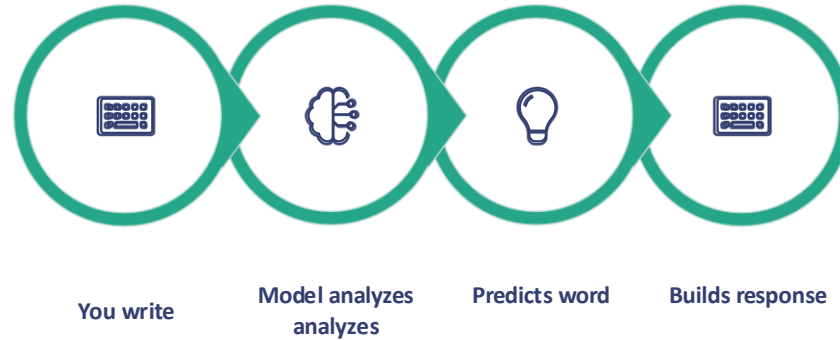
What it does: Creates brand-new content — text, images, videos, code — in response to a prompt.

Examples: ChatGPT writing a lesson plan, Gamma building a slide deck, HeyGen generating a video.



How It Works: Next-Word Prediction

At its core, a large language model like ChatGPT works by predicting the most probable next word — over and over again. This simple mechanism, applied at enormous scale, produces surprisingly fluent and useful responses.



Critical mindset: AI has **no real understanding** of what it says. It is generating statistically likely sequences of words. This is why it can sound confident and still be wrong — which brings us to our next slide.

Limitations to Know From Day One

Building a healthy, critical relationship with AI starts here. These limitations are not reasons to avoid AI — they are reasons to use it **thoughtfully**.

Hallucinations

AI can confidently produce information that sounds correct but is entirely fabricated — including fake citations, statistics, and events. Always verify facts from authoritative sources.

Incorrect Answers

AI can make calculation errors, misread context, or give outdated information. It is not a replacement for subject-matter expertise.

Overconfidence

Unlike a search engine that says "no results found," AI will always produce an answer — even when it should not. Treat every output as a first draft, not a final product.

AI Tools in Education: An Overview by Output

The AI tool landscape can feel overwhelming at first. The good news is that for educators, a small set of well-chosen tools covers almost every teaching need — from content creation to student engagement to research support.

TEXT GENERATION

ChatGPT, Claude, Gemini



PRESENTATIONS

Gamma



VIDEO CREATION

HeyGen, InVideo



RESEARCH

NotebookLM



THE COMPLETE GUIDE TO AI TOOLS

EVERY CATEGORY • TOP TOOLS • SMARTER WORK

TEXT & WRITING	PRESENTATIONS	IMAGES & DESIGN	VIDEO CREATION	RESEARCH & SCIENTIFIC WORK
ChatGPT Writing, brainstorming, coding, data, content creation Claude Long-form writing, analysis, academic support Quillbot Google AI assistant for writing & research Microsoft Copilot AI assistant in Microsoft 365 (Word, Excel, etc.) Grammarly Grammar, spelling, tone and clarity improvement Jenni AI Academic & research writing assistant Notion AI Notes, summaries, content generation	Gamma.app AI slide generation in seconds Civitas AI Design beautiful slides with AI Beautiful.ai Smart slide design and templates Tome AI-powered storytelling presentations PowerPoint Copilot Create slides, summaries and design with AI Prezi AI Interactive presentations with AI	Midjourney AI image generation from text DALL-E 3 OpenAI image generation Adobe Firefly Professional AI images by Adobe Cameo AI Designs, posters, infographics & more Leonardo AI High-quality game art & illustrations Midground Logos, posters & creative graphics	HeyGen AI avatars & talking videos Synthesia AI avatar videos from text Runway ML AI video editing & generation Pika Labs Text-to-video creative clips InVideo AI AI video creation made easy CapCut AI Smart editing, subtitles & effects Vec (Google) Advanced AI video generation	Elicit Find papers & extract data from research Scite Citation analysis & smart references Consensus AI answers from scientific papers Research Rabbit Discover & organize research papers Bulldog Read, understand & summarize papers Connected Papers Visualize research connections Semantic Scholar AI Find & analyze academic research
TEACHING & EDUCATION	GOING & PRODUCTIVITY	AUDIO & VOICE	DATA ANALYSIS	MEETINGS & PRODUCTIVITY
NotewiseLM Summarize texts, create study guides Kahoot! AI Create quizzes & interactive games Quizlet AI AI quiz generation for any topic Wayground Engaging lessons & activities MagnificSchool AI Lesson plans, rubrics, student support Curpled Interactive lessons & presentations	GitHub Copilot AI pair programmer in your IDE Cursor AI AI-first code editor Replit AI Code, design & deploy with AI Codium Free AI coding assistant ChatGPT Code generation, debugging, explanations	ElevenLabs Voice cloning & realistic speech Suno AI AI music generation Udio Create songs with AI Murf AI AI voiceovers & narration Descript Audio & podcast editing with AI	ChatGPT (Advanced) Data Analysis Analyze data, create charts, insights Tableau AI Smart data visualization and analytics Power BI Copilot AI insights in Power BI Julius AI AI data analyst & visualization	Otter.ai Meeting transcription & summaries Fireflies.ai Record, transcribe & summarize meetings Motion AI scheduling & productivity Notion AI Notes, tasks & automation
WEBSITE & APP CREATION	GAMING & INTERACTIVE AI	3D & ANIMATION	TRANSLATION & LANGUAGES	CONTENT DETECTION & SAFETY
Wix AI Build websites with AI Framer AI AI website builder Lexical Build apps & websites with natural language Bolt.new Full-stack apps in seconds Webflow AI AI site generation & design	Inworld AI AI NPCs & character conversations Character.AI AI game design & content generation Lexandria AI Game assets & textures Unity AI Tools AI assistance in game development Roblox AI Create & power experiences	Blender AI Tools AI-assisted 3D modeling, animating & rendering Sora (OpenAI) AI video generation from text Meshy AI Text-to-3D models in seconds Kaedim AI 3D model generation for games	DeepL High-quality translations Google Translate AI Translate text, speech & images ChatGPT Language learning, translation & practice	Originality.ai AI content detection & plagiarism check Turnitin Academic integrity & plagiarism check ZeroGPT Detect AI-generated content Hugging Face Detectors Open source content safety tools

GAMIFICATION & ENGAGEMENT

Make learning, training & engagement fun and effective

Lovable
Build interactive apps, courses & tools
Gemini
Create interactive presentations, infographics & games

Kahoot!
AI-powered games, games & polls
Wayground
Create engaging lessons, games & activities

★ MOST POPULAR ALL-AROUND AI TOOLS

ChatGPT
General-purpose AI assistant for writing
NotewiseLM
Summarize & research papers
Research Rabbit
Research paper discovery
Runway ML
AI-powered image generation
Midjourney
AI image generation
GitHub Copilot
AI code generation

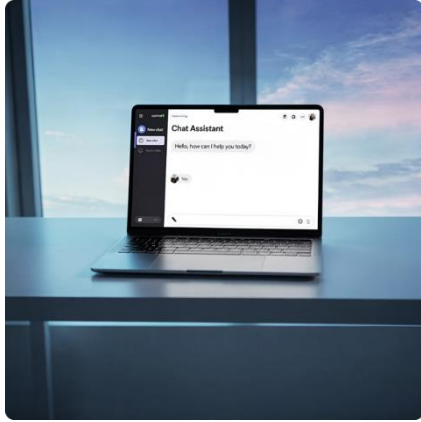
Gamma.app
AI presentation generation
Civitas AI
AI slide design for research
Beautiful.ai
AI-powered storytelling presentations
Tome
AI-powered storytelling presentations
PowerPoint Copilot
AI slide generation
Prezi AI
AI presentation generation

STRONG WORKFLOW FOR UNIVERSITY TEACHING

ChatGPT
AI assistant for writing
Gamma.app
AI presentation generation
Civitas AI
AI slide design for research
Beautiful.ai
AI-powered storytelling presentations
Tome
AI-powered storytelling presentations
PowerPoint Copilot
AI slide generation
Prezi AI
AI presentation generation

NotewiseLM
Summarize & research papers
Research Rabbit
Research paper discovery
Runway ML
AI-powered image generation
Midjourney
AI image generation
GitHub Copilot
AI code generation

AI is evolving every day. Explore, learn & use responsibly! | Save time. Boost creativity. Do more with AI. | Always fact-check, protect privacy & use AI ethically.



ChatGPT

By OpenAI. The most widely used AI assistant. Excellent for generating lesson plans, explanations, quiz questions, and student feedback. Free tier available.



Claude

By Anthropic. Known for longer context windows and careful, nuanced responses. Particularly strong for analysing and summarising lengthy documents.



Gemini

By Google. Deeply integrated with Google Workspace tools like Docs, Sheets, and Slides — making it a natural fit for educators already in the Google ecosystem.

Text Generation Tools

These three tools are the backbone of AI-assisted content creation for teachers. Each has its strengths — try them to find your preferred working style.

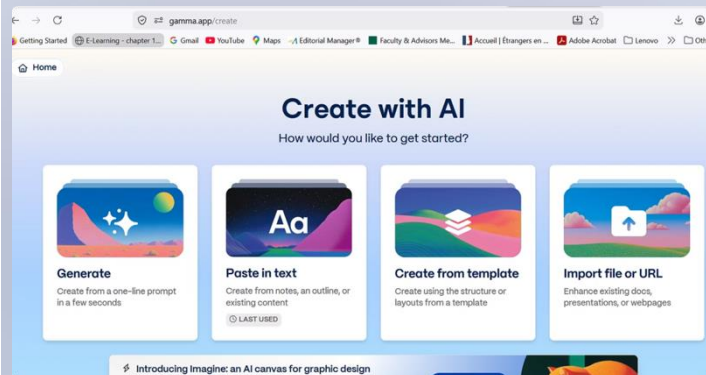
Presentation Tool: Gamma.app

What It Does

Gamma transforms plain text or a simple outline into a fully designed, visually polished slide deck — automatically. No design skills required.

Why It Matters for Educators

- Turn a ChatGPT-generated lesson outline into slides in under two minutes.
- Great for creating student-facing resource packs, revision guides, and topic overviews.
- Outputs are responsive and shareable via link — no software installation needed for viewers.
- You are literally using it right now in this session.



Video Generation Tool: [HeyGen.com](https://www.heygen.com)

HeyGen allows you to create short, professional-looking explainer videos featuring a realistic AI avatar — without a camera, microphone, or video editing skills. You simply write a script, choose an avatar, and the video is generated for you.

Flipped Classroom

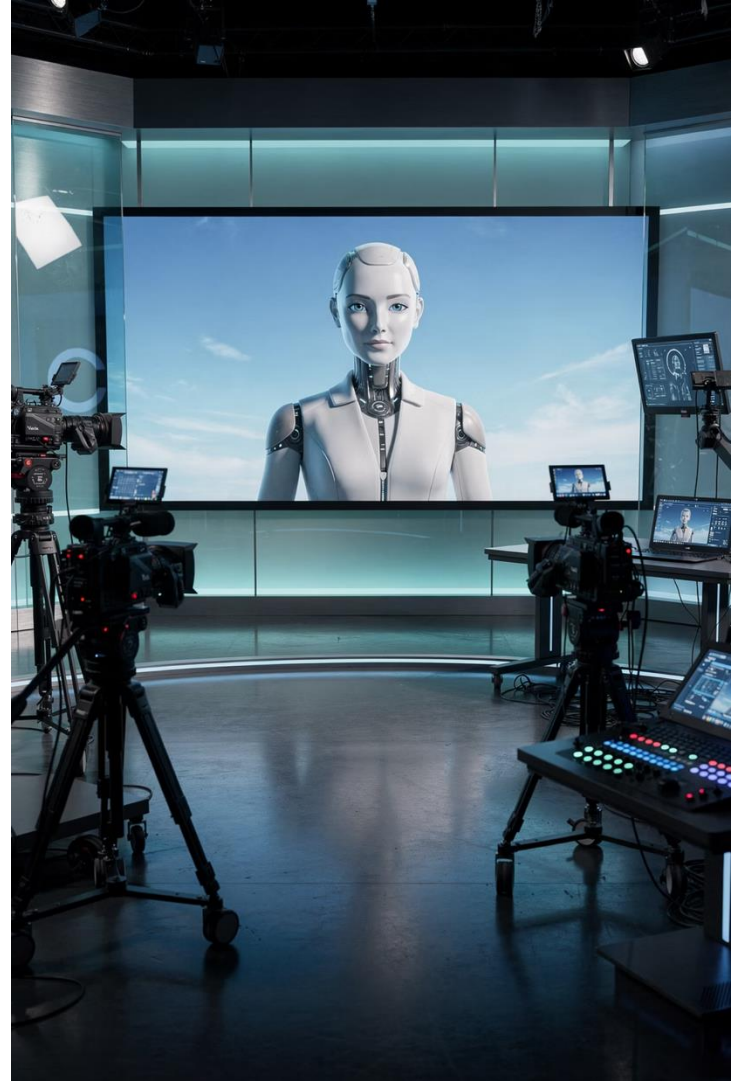
Record a concept explanation once and let students watch it before class, freeing up lesson time for discussion and practice.

Multilingual Content

HeyGen supports multiple languages, making it valuable for diverse student cohorts or international programmes.

Accessibility

Auto-generated subtitles and the ability to re-watch content supports students with different learning needs.



Video Generation Tool: InVideo.ai

InVideo allows you to create professional-quality videos from text prompts or scripts — no camera, editing software, or technical skills required. Simply describe your video or paste a script, choose a style, and InVideo generates a fully edited video with visuals, music, and voiceover.

Course Trailers

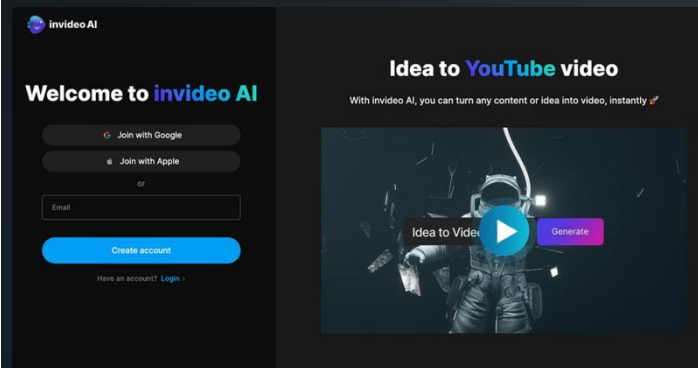
Create engaging promotional videos for your modules or courses to boost student motivation and enrolment.

Concept Explainers

Turn complex topics into short, visually rich videos that students can revisit at their own pace.

Social Media Content

Repurpose your teaching content into short clips for LinkedIn, Instagram, or institutional channels.



Research Tool: [Notebooklm.google.com](https://notebooklm.google.com)

What Makes It Different?

NotebookLM, by Google, is a research assistant that works **exclusively with the documents you upload**. Unlike ChatGPT, it does not hallucinate facts from the broader internet — it only references your uploaded sources.

This makes it significantly more reliable for academic and curriculum-related tasks where accuracy is critical.

Best Uses for Educators

- Summarise lengthy curriculum documents or policy papers.
- Generate Q&A pairs from a textbook chapter.
- Create study guides from your own course materials.
- Ask questions directly about a document and get cited answers.



What Can AI Actually Generate for You?

One of the most practical things about generative AI is the sheer variety of teaching materials it can help you create. Here is a snapshot of what is possible — all from a simple text prompt.



Explanations

Clear, level-appropriate explanations of complex concepts — adapted for beginners or advanced students.



Lesson Plans

Full structured lesson plans including objectives, timing, activities, and assessment criteria.



Exercises

Practice questions, problem sets, and scenario-based tasks tailored to your topic and learning objectives.



Slides

Complete slide decks from an outline, ready to edit and present — like this one.



Quizzes

Multiple choice, true/false, short answer — instantly generated and easily customised.

AI: THE TEACHER'S ASSISTANT, NOT THE TEACHER

WHAT AI CAN DO (CAPABILITIES)

[Draft Content Rapidly]

- CREATE LESSON PLANS
- DRAFT EMAILS
- MAKE STUDY GUIDES

Examples: Write a 10-page report

[Suggest Ideas & Creativity]

- BRAINSTORM PROJECT IDEAS
- FIND NEW EXAMPLES
- OFFER TEACHING STRATEGIES

Example: Generate five unique prompts

[Adapt Materials & Formatting]

- SIMPLIFY TEXT for different levels
- TRANSLATE content
- FORMAT data tables

Examples: Convert a reading to grade level

[Save Preparation Time]

- AUTOMATE repetitive grading
- QUICKLY find resources
- ORGANIZE materials

Examples: Reduce time from hours to minutes

WHAT AI CANNOT DO (LIMITATIONS)

THE TEACHER: THE IRREPLACEABLE HUMAN ELEMENT

[Know Your Students]

- UNDERSTAND student backgrounds
- IDENTIFY specific needs
- RECOGNIZE individual strengths

Example: True context and feeling

[Build Relationships]

- CREATE genuine empathy
- SUPPORT emotional well-being
- BUILD TRUST through time

Examples: Emotional nuances

[Verify Its Own Accuracy]

- SELF-CORRECT errors
- ENSURE total reliability
- MAKE professional ethics judgments

Example: Genuine World Knowledge

[Make Ethical Judgements]

- GUIDE moral behavior
- HANDLE sensitive situations
- ADHERE to professional codes

Example: Genuine Originality

HUMAN GUIDANCE IS ESSENTIAL

• ETHICAL FRAMEWORKS

• STUDENT-CENTERED FOCUS



Live Demonstration

From Idea to Teaching Material in 5 Minutes

We are now going to watch the full workflow live. In four steps, using real tools, we will take a single teaching idea and transform it into a complete set of learning materials — a lesson plan, slides, a short video, and a summarised research document.

1

ChatGPT

Generate a structured lesson plan

2

Gamma

Convert the plan into slides

3

HeyGen, InVideo

Create a short explanatory video

4

NotebookLM

Summarise supporting documents

Demo Step 1 — ChatGPT: Generate a Lesson Plan Plan

The Prompt

Type the following into ChatGPT:

"Create a structured lesson plan on soil compaction for a group of second-year civil engineering students. Include learning objectives, a 5-minute warm-up activity, a 20-minute explanation section, a 10-minute group exercise, and a closing reflection."

What to Notice

- How quickly a full, usable structure appears.
- The level of detail in the objectives and activities.
- How the output changes when you add or remove context.
- This is your first glimpse at how **prompt quality** drives output quality.

Demo Step 2 — Gamma: Generate Slides Automatically

Copy the lesson plan output from ChatGPT and paste it directly into Gamma. Within moments, Gamma will produce a fully designed slide deck — with layouts, visual hierarchy, and colour — based entirely on your text. Watch how the structure ChatGPT created translates directly into a professional presentation with zero manual design work.



This two-step workflow — ChatGPT for content, Gamma for design — is one of the most time-saving combinations available to educators today.

Demo Step 3 a — HeyGen: Generate an Explanatory Video

What We Are Doing

Using one section of the lesson plan as a script, we will generate a short AI avatar video in HeyGen. The avatar speaks the script naturally, with realistic lip-sync and expression.

If a live demo is not possible in the time available, a pre-recorded screenshot walkthrough can be shown instead.

Teaching Applications

- Create a "concept introduction" video for students to watch before class.
- Record a welcome message for an online module.
- Produce revision summary videos at the end of a unit.
- All without appearing on camera yourself, if preferred.

Demo Step 3b — InVideo: Generate a Video from a Script

What We Are Doing

Using the lesson plan or script generated in Step 1, we will create a short video in InVideo. Simply paste the script into InVideo, choose a video style and template, and the platform generates a fully edited video with visuals, transitions, music, and voiceover.

If a live demo is not possible in the time available, a pre-recorded screenshot walkthrough can be shown instead.

Teaching Applications

- Create a short course trailer to introduce a new module to students.
- Turn a written explanation into a visual video summary for revision.
- Produce social media clips to share learning content beyond the classroom.
- Generate multilingual video content by switching the voiceover language.



Demo Step 4 — NotebookLM: Summarize Content

Upload a relevant academic paper, curriculum document, or textbook chapter to NotebookLM. Ask it to summarise the key points, generate five discussion questions, or identify the main concepts relevant to your lesson topic.



Upload your source documents

PDFs, Google Docs, or web links — NotebookLM reads and indexes them for you.



Ask targeted questions

Query the document directly: "What are the main causes of soil compaction according to this paper?"



Get cited, grounded answers

Every response links back to the source — far more reliable than a generic AI search.

Reflection: What Did You Notice?

Before we move into your hands-on practice, take a moment to reflect on the demonstration. There are no wrong answers here — your observations are valuable for the whole group.



What Seemed Useful?

Which part of the workflow would most immediately save you time or improve a material you currently create manually?



What Seemed Missing?

Where did the AI output feel incomplete, generic, or not quite right for your specific teaching context?



What Felt Risky?

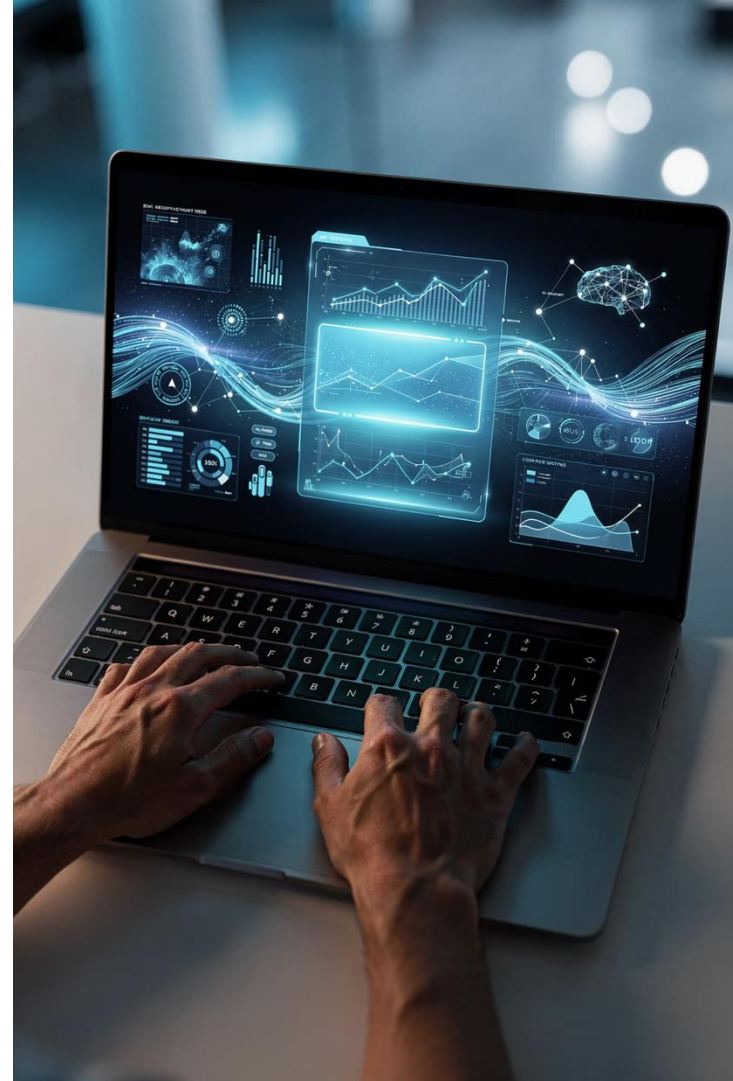
What concerns do you have about accuracy, student use, academic integrity, or over-reliance on AI tools?

See 4 11 5 Y u e 0 B 1 5 e p z q 5 x x 1 z ° u l e p 9 7 0 7 1 9 K 6 7 p x e 7 e 1 ſ w
3 Ẓ T 3 ů C T ů 4 9 3 Ÿ ž 7 ˘



Now It's Your Turn

You have watched the workflow. You have seen the tools. Now you will experience it yourself — writing your first AI prompt and generating your first piece of teaching content.



Activity: Create Your First Prompt

Task: Generate a Course Handout

Open ChatGPT (or Claude or Gemini — your choice). Your goal is to write a prompt that generates a useful, structured course handout for one of your own teaching topics. Use the template below as a guide.

"Create a one-page course handout on **[your topic]** for **[your student level]**. Include a brief introduction, 3–4 key concepts with explanations, a summary box, and 2 reflection questions for students."

- ❑ Do not worry about getting the perfect prompt on the first try. Write one, review the output, and then refine it. That process of iteration **is** the skill we are building.

Activity Guidelines: What to Include in Your Prompt

The quality of your AI output is directly proportional to the quality of the information you provide. A vague prompt produces vague results. Think of writing a prompt like briefing a very capable but very new colleague.

1

Topic

Be specific. Instead of "science," write "the water cycle for Year 8 geography." The more precise, the more useful the output.

2

Student Level

Specify the year group, course level, or prior knowledge. "Introductory undergraduate" produces very different content from "advanced A-level."

3

Structure

Tell the AI what format you want: a table, bullet points, numbered steps, a two-column layout. Explicit structure requests produce cleaner, more usable outputs.

Optional Extension: Generate Slides with Gamma

If You Finish Early

Take the handout text your AI just generated and paste it into Gamma. In the prompt box, type: **"Create a slide deck from this content."** Watch the transformation happen in real time.

Things to Try

- Ask Gamma to add more visual variety to the deck.
- Edit a slide manually to see how easy the interface is.
- Change the theme colour to match your institution's branding.
- Share the link with a neighbour and ask for their feedback.

- ✔ This two-tool workflow — ChatGPT to Gamma — takes under five minutes once you are comfortable with it. Many educators now use it as a standard part of their lesson prep.

Compare and Reflect on Your Outputs

Now that you have generated your first AI content, take a few minutes to evaluate what you received. Critical evaluation of AI output is a core professional skill — not an optional extra.



Quality

Is the language appropriate for your student level? Is the tone professional? Does it read like something you would actually use, or does it need significant editing?



Accuracy

Are the facts, concepts, and terminology correct? Would a student following this content learn the right things? Always verify subject-specific claims independently.



Usability

Could you use this output in your class next week — with minimal editing? What would you need to change, add, or remove to make it truly your own?



Key Takeaways from Session 1

AI Is Powerful — But Imperfect

Hallucinations, incorrect facts, and overconfident responses are real limitations. Every AI output needs a human review before it reaches students.

Prompts Matter More Than the Tool

The same AI tool will produce dramatically different results depending on how you phrase your request. Precision, context, and structure in your prompt drive quality in your output.

AI Supports Teaching — It Does Not Replace It

Your professional judgement, knowledge of your students, and ability to build relationships are irreplaceable. AI handles the drafts; you bring the expertise and humanity.

Coming Up: Session 2

Prompt Engineering

In Session 2, we go deeper into the single most important skill for working with AI effectively — **writing great prompts**. You will learn the frameworks, patterns, and techniques that transform average AI outputs into genuinely useful teaching materials.

1

Session 1 

Introduction to AI & Content Generation — completed today.

2

Session 2 →

Prompt Engineering — writing prompts that deliver results.

3

Session 3

AI as a Teaching and Learning Assistant

4

Session 4

Ethical Issues, Responsibilities, and Learning Designer

